Bishop - Gromov volume comparison

Let M be an n-dim Riem. mfld with $Ric \geq (n-1)k$ $\quad k \in \mathbb{R}$

Let $v(n,k,r)$ = volume of r-ball in n-dim space form with constant curvature k

Then for $x \in M$ $\quad \dfrac{vol(B(x,r))}{v(n,k,r)}$ is decreasing with r

(limit $= 1$ as $r \to 0$)

<u>Theorem</u> For $n \geq 2$, $k \in \mathbb{R}$, $D > 0$ the following classes are precompact in $(\mathcal{M}, d_{GH})$

(1) Riem. n-folds with $Ric \geq (n-1)k$ and $Diam < D$

(2) pointed complete Riem n-folds with $Ric \geq (n-1)k$

<u>Proof</u> (1) Let $\varepsilon > 0$. Need bound $N(\varepsilon)$ on number of disjoint ε-balls in M^n with $Ric \geq (n-1)k$, $diam < D$.

By volume comparison for $x \in M$ we have

(for $\varepsilon < D$) $$\frac{vol(B(x,\varepsilon))}{v(n,k,\varepsilon)} \geq \frac{vol(B(x,D))}{v(n,k,D)}$$

So $\quad \mathrm{Vol}(B(x,\varepsilon)) \geq \dfrac{v(n,k,\varepsilon)}{v(n,k,D)} \underbrace{\mathrm{vol}(B(x,D))}_{\mathrm{vol}(M)}$

$\Rightarrow$ if $B(x_1,\varepsilon), \dots, B(x_k,\varepsilon)$ are disjoint, then $k \leq \dfrac{v(n,k,D)}{v(n,k,\varepsilon)} = N(\varepsilon)$

$\square$

(2) is similar.

(Recall continuity method for finding positive KE metrics : $\mathrm{Ric}(\omega_t) = t\omega_t + (1-t)\omega$
$\mathrm{Ric}(\omega_t) \geq t\omega_t$)

Convergence of Riemannian manifolds

<u>Definition</u> Let (M_i, g_i, p_i), (M, g, p) be complete pointed Riem. mflds. We say $(M_i, g_i, p_i) \longrightarrow (M, g, p)$ in the pointed $C^{m,\alpha}$ topology ($m \geq 0$, $\alpha \in (0,1)$) if for all $R > 0$ there is an open $\Omega \supset B(p,R)$ and embeddings $F_i : (\Omega, p) \hookrightarrow (M_i, p_i)$ for large i s.t. $B(p_i, R) \subset F_i(\Omega)$ and $F_i^* g_i \to g$ in $C^{m,\alpha}(\Omega)$

<u>Note</u>: 1. In this case
$(M_i, g_i, p_i) \to (M, g, p)$ in the pointed GH-sense (HW)

2. If (M_i, g_i) have uniformly bounded diameter then for large i M_i and M must be diffeomorphic. In particular if a class $\mathcal{C}$ of manifolds is pre-compact in $C^{m,\alpha}$ and has diam $< D$, then $\mathcal{C}$ has finitely many diffeomorphism types.

<u>Q</u>. What classes $\mathcal{C}$ are precompact in $C^{m,\alpha}$?

<u>Norms of Riemannian manifolds</u>

<u>Defn</u> Let (M, g, p) be a pointed Riem n-fold. Define $C^{m,\alpha}$-norm at p on scale $r > 0$ as follows:

$$\|(M, g, p)\|_{C^{m,\alpha}, r} \leq Q \text{ means}$$

$$\exists \text{ a } C^{m+1,\alpha}\text{-chart } \varphi : (\underset{\mathbb{R}^n}{\underset{\cap}{B(0,r)}}, 0) \to (\underset{M}{\underset{\cap}{U}}, p)$$

such that (1) $|D\varphi| \le e^{Q}$ on $B(0,r)$

$|D\varphi^{-1}| \le e^{Q}$ on U

i.e. in chart components of g satisfy $e^{-2Q} f_{ij} \le g_{ij} \le e^{2Q} f_{ij}$

(2) For all multiindices I, $0 \le |I| \le m$

$$r^{|I|+\alpha} |\partial^I g_{ke}|_{C^\alpha} \le Q$$

Define $\|(M,g)\|_{C^{m,\alpha},r} = \sup_{p \in M} \|(M,g,p)\|_{C^{m,\alpha},r}$

<u>E.g.</u> • $\|(\mathbb{R}^n, g_{Euc})\|_{C^{m,\alpha},r} = 0$ for all $r > 0$

(this characterizes Euclidean space)

• If (M,g) is flat, then

$\|(M,g)\|_{C^{m,\alpha},r} = 0$ for $r < \mathrm{inj}(M,g)$

<u>Basic properties of the norm</u>

1) $\|(M,g,p)\|_{C^{m,\alpha},r} = \|(M,\lambda^2 g,p)\|_{C^{m,\alpha},\lambda r}$ for all $\lambda > 0$.

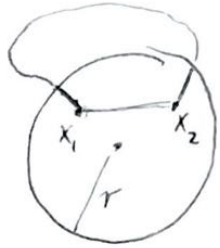
2) $r \longmapsto \|(M,g,p)\|_{C^{m,\alpha},r}$ is increasing, continuous, has limit 0 as $r \to 0$

3) If $(M_i, g_i, p_i) \to (M, g, p)$ in $C^{m,\alpha}$

then $\|(M_i, g_i, p_i)\|_{C^{m,\alpha},r} \to \|(M, g, p)\|_{C^{m,\alpha},r}$ $\forall r > 0$

4) If $\|(M, g, p)\|_{C^{m,\alpha},r} < Q$ is realized by chart φ, then for all $x_1, x_2 \in B(0,r)$

$$e^{-Q} \min\{|x_1 - x_2|, 2r - |x_1| - |x_2|\} \leq d(\varphi(x_1), \varphi(x_2)) \leq e^{Q} |x_1 - x_2|$$



5) $\|(M, g, p)\|_{C^{m,\alpha},r}$ is realized by a $C^{m+1,\alpha}$-chart.